



CNT140AB Introduction to Networks
Fall 2024-Section 28346

Course Information

Instructor: Wayne Kibbe

Email: wayne.kibbe@paradisevalley.edu

Office Hours: Email for appointment

Class Dates: 8/26-11/20

Format: Online

Credits: 4

Final Exam Time: 11/20 until 11:59 pm

Students must take the final exam to earn credit for this course regardless of their grade.

Course Description:

Students will focus on the architecture, structure, functions, components, and models of the Internet and other computer networks. Principles and structure of IP addressing and fundamentals of Ethernet concepts, media, and operations are introduced. Students will build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes. This course prepares students for the Cisco certification examination. **Prerequisites:** None.

Supplies:

- pen or pencil
- notebook
- USB flash drive, external hard drive, or Google Drive

Student-generated data is not stored on classroom computers; therefore, students must use their storage. Proper data storage is critical for success in this class.

Textbook: (recommended, not required)

- **Title:** Introduction to Networks Companion Guide (CCNAv7)
- **Author:** Cisco Networking Academy
- **ISBN-13:** 978-0-13-663366-2

Course Objectives:

Upon completion of this course, students should be able to:

- Explain the advances in modern network technologies.
- Implement initial settings including passwords, IP addressing, and default gateway parameters on a network switch and end devices.
- Explain how network protocols enable devices to access local and remote network resources.
- Explain how physical layer protocols, services, and network media support communications across data networks.
- Calculate numbers between decimal, binary, and hexadecimal systems.
- Explain how media access control in the data link layer supports communications across networks.
- Explain how Ethernet operates in a switched network.
- Explain how routers use network layer protocols and services to enable end-to-end connectivity.
- Explain how ARP and ND enable communication on a local area network.
- Implement initial settings on a router and end devices.
- Calculate an IPv4 subnetting scheme to efficiently segment your network.
- Implement an IPv6 addressing scheme.
- Use various tools to test network connectivity.
- Compare the operation of transport layer protocols in supporting end-to-end communication.
- Explain the operation of the application layer in providing support to end-user applications.
- Configure switches and routers with device hardening features to enhance security.
- Troubleshoot connectivity in a small network.

Time Commitment:

Students should expect to spend 16 or more hours per week working on this course. Hours may vary based on the student's skill levels.

Web-Based Third-Party Software:

This course uses Cisco's netacad.com as a web-based third-party tool. This tool will be used to complete course examinations and quizzes along with accessing the course reading materials. Students may be required to submit work and/or download information from these tools; therefore, there is some risk that individuals electing to use the products and services made available by these tools may place any student information shared with the tool vendor at risk of disclosure.

Netacad.com & Packet Tracer:

<https://www.netacad.com/>

Terms of Use:

https://www.netacad.com/sites/default/files/terms/cisco_networking_academy_terms_and_conditions_july_2016_v3_english.pdf

Privacy Policy:

<https://www.cisco.com/c/en/us/about/legal/privacy.html>

Accessibility Statement:

<https://www.netacad.com/accessibility>

Wireshark:

<https://www.wireshark.org/>

Terms of Use:

<https://www.gnu.org/licenses/old-licenses/gpl-2.0.html>

TeraTerm:

<https://osdn.net/projects/ttssh2/downloads/72009/teraterm-4.105.exe/>

Grading:

Grading Scale		Weighting Rubric	
A	90 – 100%	Weekly Assignments	30%
B	80 – 89%	Discussions	10%
C	70 – 79%	Exams	20%
D	60 – 69%	Skills Based Assessment	20%
F	<60%	Final Exam	20%
		Total	100%

Course Schedule:

Week	Assignment
Week 1 8/26 – 8/31	Syllabus Acknowledgement: 5pts Week 1 Assignments: 100pts
Week 2 9/1 – 9/7	Read Modules 1-3 Week 2 Assignments: 100pts
Week 3 9/8 – 9/14	Module 1-3 Exam: 100pts Week 3 Assignments: 100pts Read Modules 4-7
Week 4 9/15 – 9/21	Module 4-7 Exam: 100pts Read Modules 8-10 Week 4 Assignments: 100pts
Week 5 9/22 – 9/28	Module 8-10 Exam: 100pts Read Modules 11-13 Week 5 Assignments: 100pts
Week 6 9/29 – 10/5	Week 6 Assignments: 100pts
Week 7 10/6 – 10/12	Week 7 Assignments: 100pts
Week 8 10/13 – 10/19	Module 11-13 Exam: 100pts Read Modules 14-15 Week 8 Assignments: 100pts
Week 9 10/20 – 10/26	Week 9 Assignments: 100pts
Week 10 10/27 – 11/2	Module 14-15 Exam: 100pts Read Modules 16-17 Week 10 Assignments: 100pts
Week 11 11/3 – 11/9	Module 16-17 Exam: 100pts Week 11 Assignments: 100pts
Week 12 11/10 – 11/16	Skills Based Assessment: 100pts Course Feedback: (prerequisite for taking the final) Final Exam: 100pts

The schedule may change during the semester. Changes will be announced in class and posted in the PVCC Canvas Shell.

Late Work Policy:

Late Labs and Packet Tracer activities will automatically receive 50% of the credit earned for the assignment.

Missed exams will receive a grade of zero (0).

Please do not copy and paste answers from the instructor's lab manuals or other resources as this is the easiest and fastest way to get zero (0) points on an assignment.

Remote Classroom Etiquette:

During remote instruction or while using telecommunications platforms (Webex, Zoom, Google Meet, etc.), remember to:

- Always present yourself professionally, just as you would in a classroom.
- Conduct yourself in all chat rooms and on discussion posts with proper decorum.
- Make sure the workspace is free of excess noise.
- Mute your microphone when you are not speaking.
- Ensure your background is suitable, especially when the camera is on.
- Be on time and pay attention to the presenter.
- Maintain the privacy of any class recordings. In other words, no sharing beyond the classroom learning platform.
- Always monitor video, microphone, and screen sharing settings to ensure only appropriate content is broadcast through the platform.

Paradise Valley Community College (PVCC) and Maricopa County Community College District (MCCCD) Policies:**Classroom Conduct Policy:**

You are expected to treat your instructor and your classmates with respect. In all correspondence, whether communicating in person, virtually, or online, you should show respect for the viewpoints of others who may disagree with you or see things from a different perspective. Criticizing, ridiculing, insulting, or belittling others will not be accepted. Avoid coarse, rough, or rude language and sarcasm. Keep in mind that electronic communications do not have the advantage of nonverbal cues that are so much a part of interpersonal

communication. Humor or satire can sometimes be misinterpreted in strictly electronic communication forums. Observe good grammar and spelling.

If the instructor determines that the student's conduct is not consistent with the classroom conduct policy, the student may be removed from the discussion forum, virtual lecture or classroom, or dropped from the course entirely.

For Your Protection:

It is your responsibility to SAVE your assignments in a folder, on your personal computer, or another removable media. Keep these copies in a safe place until you have received your final grade.

Withdrawal Policy:

The Business/IT Division recommended policy states that instructors may withdraw students who accumulate unofficial absences in excess of the number of times a class meets per week. Online students who do not participate, (e.g., fail to turn in assignments) will be withdrawn. Students may withdraw themselves without an instructor's signature (see date in my.maricopa.edu). If a student requests a withdrawal after the self-withdraw period, the instructor may withdraw a student with a Y if the cumulative grade is failing at that point.

Attendance:

Only persons who are registered for the class may attend that class. Attendance requirements are determined by the course instructor. Students who do not meet the attendance requirement, as determined by the course instructor, may be withdrawn.

Students who fail to attend the first scheduled class meeting, or to contact the instructor regarding absence before the first scheduled class meeting may, at the option of the instructor, be withdrawn.

Online students who do not post in Canvas the first week of class risk being withdrawn from the class. Posting in Canvas the first week is equivalent to showing up on the first day of class.

At the beginning of each course, each faculty member will provide students with written attendance requirements. It is the student's responsibility to consult

with the instructor regarding official or unofficial absences. Students bear the responsibility of notifying the Office of Admissions and Records when they discontinue studies in a course or at the college (refer to the Withdrawal Procedures).

Student Maricopa Email Policy:

Your MEID@maricopa.edu email account is the official form of communication between PVCC and you. You are responsible for reading college messages and responding accordingly. You should check your my.maricopa.edu account for official college/district messages and information regarding tuition and fees.

Official Absences:

MCCCD policy states that: official absences are those that occur when students are involved in an official activity of the college, i.e., field trips, tournaments, and/or athletic events, and present an official absence excuse form. Absences for such events shall not count against the number of absences allowed by an instructor or department. Students who must miss a class for an official reason must obtain an official absence verification card from the appropriate dean or associate dean and present it to the appropriate instructor(s) before the absence. Prior arrangements must be made with each instructor for make-up work. If prior arrangements have been made, the students will not be penalized.

Other official absences include jury duty and subpoenas. Appropriate documentation will be required. Prior arrangements must be made with each instructor for make-up work. If prior arrangements have been made, the students will not be penalized.

In the event of the death of an immediate family member, absences for periods of up to one week will not be counted against the number of absences allowed by an instructor or department. Students should contact the instructor(s) as soon as possible to arrange for make-up work. Appropriate documentation will be required, e.g., a copy of the obituary or funeral program. In specialized programs that require clinical rotations, this regulation may not apply.

Religious Holidays:

MCCCD policy states that: students shall have the right to observe major

religious holidays without penalty or reprisal by any administrator, faculty member, or employee of the Maricopa Community Colleges. Absences for such holidays shall not count against the number of absences allowed by an instructor or department. At least one week before the holiday, students shall submit to their instructor(s) a written statement that includes both the date of the holiday and the reason why class attendance is impossible. Prior arrangements must be made with each instructor for make-up work. If prior arrangements have been made, the student will not be penalized.

Title IX Statement:

Addressing Incidents of Title IX Sexual Harassment

In accordance with Title IX of the Education Amendments of 1972, MCCCDC prohibits unlawful sexual harassment against any participant in its education programs or activities. Sexual harassment includes quid pro quo (this for that) harassment, hostile environment, sexual assault, dating/domestic violence, and stalking. This prohibition against sexual harassment—including sexual violence—applies to students, MCCCDC employees, and visitors to campus.

The policy of MCCCDC is to provide an educational, employment, and business environment free of sexual violence, unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct or communications constituting sexual harassment as prohibited by state and federal law. Incidents of Sexual Misconduct should be reported to the college Title IX Coordinator, as outlined in the policy. MCCCDC will provide on its [Title IX and Preventing Sexual Harassment](#) webpage a link to all [Title IX Coordinators](#) at the MCCCDC. Reports may be filed anonymously at: <https://district.maricopa.edu/consumer-information/reporting>.

Student Disabilities:

Classroom Accommodations for Students with Disabilities

In accordance with the Americans with Disabilities Act (ADA), MCCCDC and its associated colleges are committed to providing equitable access to learning opportunities to students with documented disabilities (e.g., mental health, attentional, learning, chronic health, sensory, or physical). For each class/term/semester that a student needs academic adjustments/accommodations, the qualified student is required to work with

Disability Resources and Services Office (DRS) at their individual college(s). College DRS offices may only offer accommodations/academic adjustments for their own college. Contact with DRS should be made as soon as possible to ensure academic needs are met in a reasonable time. New and returning students must request accommodations each semester through DRS Connect online services. To learn more about this online service process, please contact your local DRS office.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations, you are welcome to contact DRS by using the information listed on the following webpage: <https://district.maricopa.edu/consumer-information/disability-resources/contacts>. DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions qualifying for accommodations/academic adjustments. Reasonable accommodations are established through an interactive process between you, your faculty, and DRS; and only those academic adjustments/reasonable accommodations granted by DRS are recognized by the college and MCCC. It is the policy and practice of MCCC to create inclusive and accessible learning environments consistent with federal and state law.

Students' Rights and Responsibilities:

Students are responsible for reading and understanding the districtwide scholastic standards found in the current student handbook and are responsible for the college policies included in the college catalog.

Student Code of Conduct:

Students engaging in the following are subject to disciplinary sanctions outlined in the PVCC Catalog, Student Policies Section.

Acts of Dishonesty examples include:

- Plagiarism
- Furnishing false information
- Falsifying records related to coursework
- Forgery, alteration, or misuse of any college document
- Tampering with the election of any college-recognized official
- Disruption or obstruction of teaching, research, administration, disciplinary

proceedings, or college activities. An instructor can remove a student from class for disciplinary reasons

- Physical abuse, verbal abuse, threats, intimidation, harassment, coercion, and/or conduct that threatens the health or safety of any person
- Attempted or actual theft
- Failure to comply with the direction of college officials or law enforcement officers
- Unauthorized possession, duplication, or use of keys to any college premises
- Violation of federal, state, or local laws on college campuses or at college-sponsored activities
- Use, possession, or distribution of narcotics or other controlled substances
- Use, possession, or distribution of alcoholic beverages, or public intoxication
- Illegal or unauthorized possession of firearms, explosives, weapons, or dangerous chemicals
- Participation in a college demonstration that disrupts the normal operations of the college
- Obstruction of the free flow of pedestrian or vehicular traffic on college premises
- Conduct which is disorderly, lewd or indecent
- Theft or other abuse of computer time
- Theft or other copying of proprietary software found on PVCC computers and network devices
- Abuse of the judicial system

Policy on Cheating:

Students caught cheating will receive a zero (0) for the assignment, or an F for the semester. The student will also be referred to the Dean of Academic Affairs for probation or dismissal from the college.

Students will be notified by the instructor of any changes in course requirements or policies.

Students are responsible for the information contained in this syllabus.